

SPACE INVADER GAME

LE TUYET NHUNG
NGUYEN HA AN THANH





ABSTRACT

LET'S GO



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- 01 Purpose: Recreate Space Invaders using Java and OOP.
- 02 Tools: Java, OOP principles.
- 03 Key Outcomes: A functional game with controls, and scoring.



INTRODUCTION

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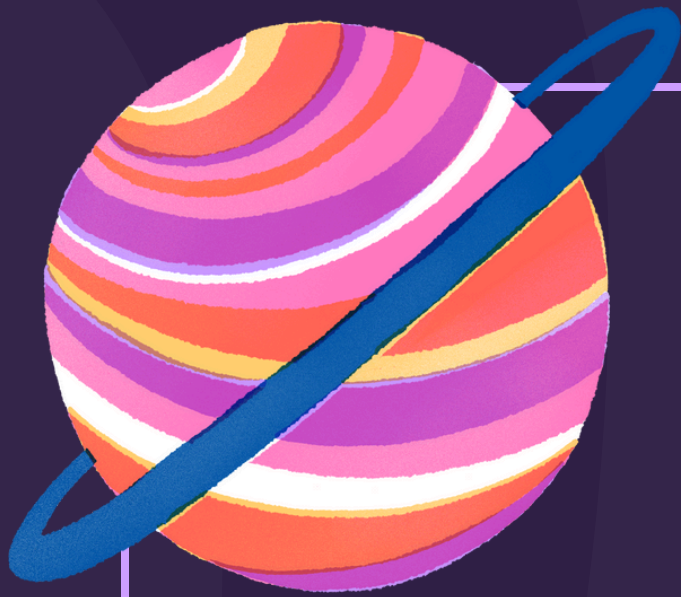
INTRODUCTION

- 01** Background: Space Invaders, released in 1978, is a pioneering arcade game that influenced the global gaming industry.
- 02** Objective: This project recreates Space Invaders using Java and OOP to improve game design and coding skills.
- 03** Structure: The game uses OOP principles, with classes for the player, aliens, shots, and game logic to ensure modularity.



REQUIREMENTS ANALYSIS

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REQUIREMENTS ANALYSIS

01

Functional Requirements

- **Player Movement:** Smooth horizontal controls for the spaceship.
- **Shooting Mechanics:** Ability to shoot bullets at aliens.
- **Enemy Waves:** Dynamic waves with increasing difficulty.
- **Scoring System:** Tracks score and unlocks levels.



REQUIREMENTS ANALYSIS

02 Non-Functional Requirements

- **Smooth Gameplay:** Ensure minimal lag for a fluid experience.
- **User Interface:** Classic, user-friendly design.
- **Platform Compatibility:** Ensure compatibility with PC.



REQUIREMENTS ANALYSIS

03 Technical Requirements

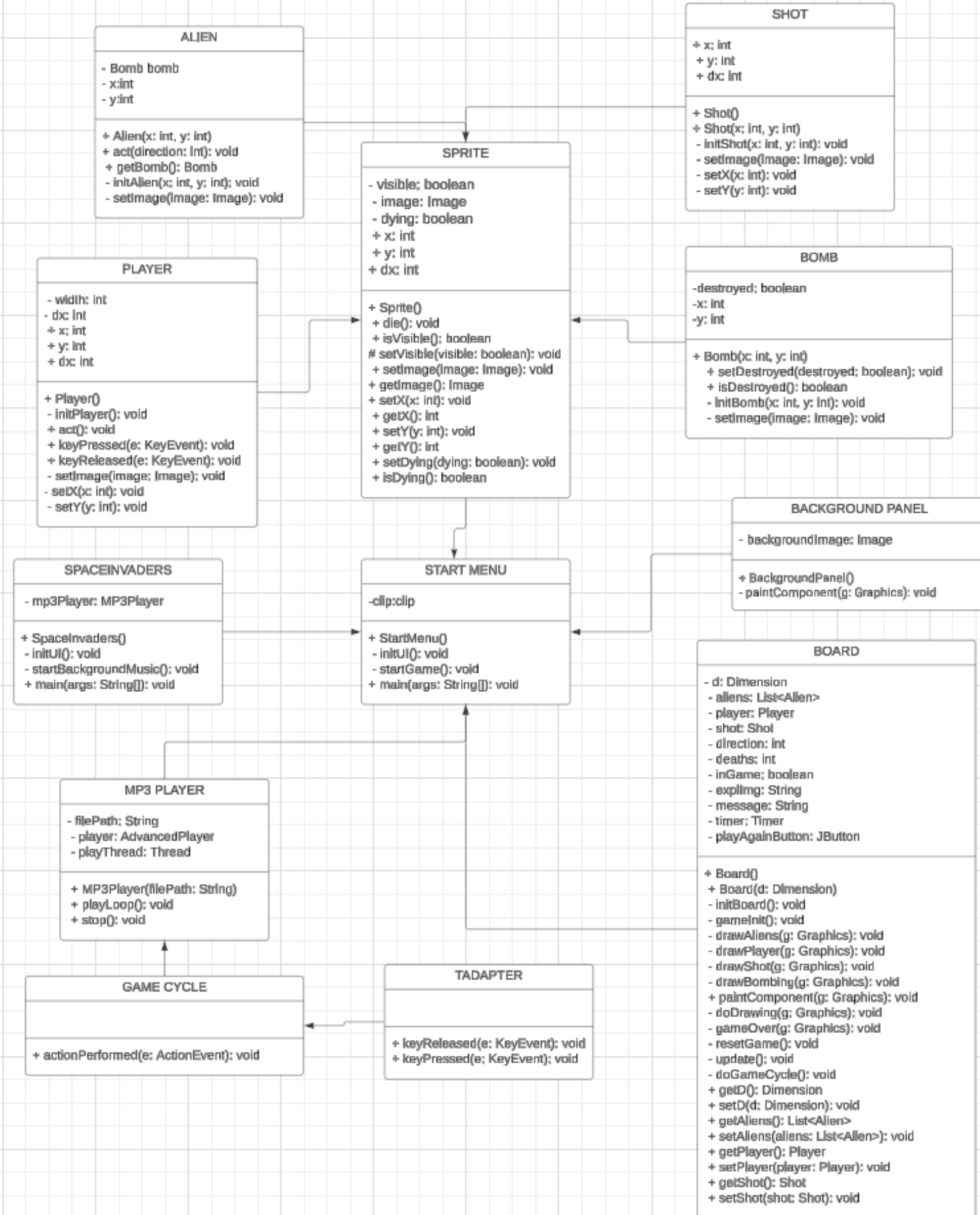
- Programming Language: Java for OOP and cross-platform support.
- Libraries and Tools: Java Swing/JavaFX for graphics and events.
- Development Tools: IDEs like NetBeans or IntelliJ IDEA for project management.



GAME DESIGN

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UML DESIGN





IMPLEMENTATION

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RESULT

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ADERS

e

C

assets

> sprite

Board.java

Commons.java

MP3Player.java

SpaceInvaders.java

StartMenu.java 3

images

alien.png

bomb.png

explosion.png

logo.png

player.png

> OUTLINE

> TIMELINE

JAVA PROJECTS

SpacInvad...

src

assets

Board

Commo... +

MP3Player

SpacInvad...

StartMenu 3

14

15

16

17

18

19

20

21

22

23

24

25

int ALIEN_WIDTH = 20;

Space Invaders

PROBLEMS

PS E:\Spa

pzmfl7gjk

PS E:\Spa

\DELL\AppData

PS E:\Spa

\DELL\AppData

Run: SpacInvaders

pData\Local\Temp\cp_9ws72t72tcotz

a\jdk-23\bin\java.exe' '@C:\Users

a\jdk-23\bin\java.exe' '@C:\Users

THANK'S FOR YOUR LISTENING

QUESTION&ANSWER

